

Trustworthy Multi-Modal Ultrasound Fusion via Uncertainty Calibration and Conflict Resolution

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Abstract—Multi-modal ultrasound combines tissue information from multiple imaging perspectives, enabling more comprehensive lesion assessment. However, conventional multi-view learning methods typically assume uniform modality quality, ignoring variability caused by imaging noise and patient-specific factors. This oversight limits diagnostic reliability, especially when some modalities provide uncertain or conflicting information. To address this, we identify two key challenges in multi-modal ultrasound fusion: 1) how to quantify modality-wise uncertainty, and 2) how to resolve conflicts among predictions. We propose a novel method, termed TMUF (Trustworthy Multi-modal Ultrasound Fusion), which dynamically integrates information from different modalities through *uncertainty calibration and conflict resolution*. Specifically, we introduce a cross-modal uncertainty calibration regularizer to estimate evidence-based uncertainty across modalities, aligning uncertainty with prediction correctness. We further develop a credibility-aware fusion strategy that evaluates cross-modal consistency and uncertainty to distinguish credible from non-credible modalities, assigning fusion weights accordingly. We validate TMUF on public and private datasets for breast lesion and liver cancer diagnosis. The proposed method achieves diagnostic accuracies of 88.00% and 92.08%, respectively, outperforming

state-of-the-art baselines. These results demonstrate the effectiveness of TMUF in enhancing diagnostic accuracy and robustness for multi-modal ultrasound.

Index Terms—Conflict evidence fusion, multi-modal ultrasound, trustworthy learning, uncertainty calibration.

I. INTRODUCTION

HEPATOCELLULAR carcinoma (HCC) and intrahepatic cholangiocarcinoma (ICC) are the two major subtypes of primary liver cancer, collectively accounting for the third leading cause of cancer-related mortality globally [1]. Owing to the lack of early symptoms, both are often diagnosed at intermediate or advanced stages, limiting therapeutic options and prognosis. Clinically, HCC and ICC require distinct treatment strategies: HCC is typically amenable to curative interventions such as transplantation, resection or ablation; whereas ICC typically necessitates systemic chemotherapy or palliative care [2]. Notably, curative resection for ICC usually demands a wider surgical margin than for HCC due to its more invasive nature [3]. Therefore, accurate differentiation between ICC and HCC is critical for optimizing treatment planning and improving clinical outcomes.

Multi-modal ultrasound has emerged as a principal imaging technique in disease screening and diagnosis, particularly with advancements in modalities such as B-mode ultrasound (BUS), contrast-enhanced ultrasound (CEUS), and elastography ultrasound (EUS) [4]. BUS captures anatomical details that help identify structural abnormalities (i.e., margins, cysts, and tissue deformations). CEUS complements BUS by visualizing tissue perfusion in real time, providing valuable insights into tumor vascularization patterns [5], [6]. By capturing contrast dynamics across enhancement phases, CEUS enables functional evaluation of microvessel distribution, tissue necrosis, and tumor invasion [7]. Elastography ultrasound further complements BUS and CEUS by quantifying tissue stiffness. However, differentiating HCC from ICC using multi-modal US remains a critical challenge. Approximately 52% of small ICC lesions in cirrhotic livers exhibit CEUS enhancement patterns resembling HCC, including global arterial hyperenhancement and washout—features traditionally associated with HCC [8]. This overlap is particularly common in nodules ≤ 5 cm, contributing to high misdiagnosis rates. Moreover, around 60% of ICC cases show inconsistent vascular patterns between conventional ultrasound and CEUS, further complicating interpretation and exacerbating inter-observer variability. These challenges underscore the need for automated diagnostic

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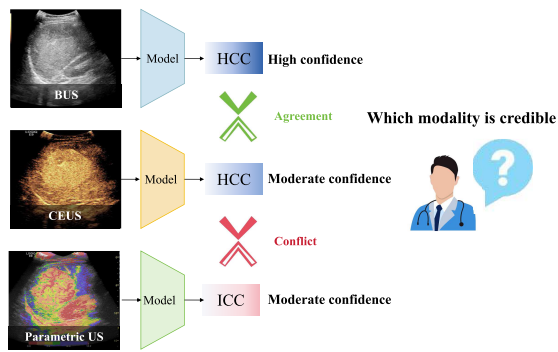


Fig. 1. An illustration of multi-modal ultrasound fusion in liver cancer diagnosis. In this example, both BUS and CEUS modalities consistently predict HCC, albeit with different confidence levels. In contrast, the parametric ultrasound view yields a contradictory prediction of ICC. It raises a fundamental question: Which modality should be considered credible for clinical decision-making?

systems capable of fusing multi-modal ultrasound, providing a more comprehensive and nuanced evaluation of lesions.

Typically, most existing methods rely on the assumption that multi-modal inputs possess stable quality and aligned semantic content [9], [10], which may not hold in real-world clinical environments. As illustrated in Fig. 1, both B-mode ultrasound (BUS) and contrast-enhanced ultrasound (CEUS) suggest a hepatocellular carcinoma (HCC) diagnosis, albeit with varying confidence. In contrast, the parametric ultrasound view, in which each pixel encodes the arrival time of contrast agents, detailed in Section IV-A produces a conflicting prediction of intrahepatic cholangiocarcinoma (ICC), demonstrating a clear semantic conflict across modalities. Such conflicts may stem from intrinsic modality discrepancies, variations in imaging quality, or operator-dependent factors. However, the subject- and modality-specific reliability is often overlooked in prior works [11], [12]. Recent theoretical studies [13] emphasize the importance of dynamic fusion strategies that adapt to modality quality and selectively suppress misleading modalities. This raises a central question in multi-modal learning [14]: *Which modality is credible?* To this end, we propose an uncertainty-aware dynamic fusion strategy that quantitatively evaluates the credibility of each input modality for trustworthy decision-making.

In the context of medical ultrasound, modality credibility associates with two types of uncertainty: 1) aleatoric uncertainty and 2) epistemic uncertainty [15], [16]. Aleatoric uncertainty reflects inherent randomness in data, caused by imaging noise, inconsistent plane selection, and uncontrolled patient movement, etc. As it is tied to the stochastic nature of clinical US, it is irreducible even with more data. In contrast, epistemic uncertainty stems from limited training data, missing modalities, or model incapacity to represent the full data distribution. This brings us to the first fundamental question: *How to quantify the uncertainty in multi-modal US?* Existing uncertainty modeling techniques include Bayesian neural networks (BNNs), deep ensembles, and evidential deep learning. BNNs [17] represent model parameters as probability distributions, while ensemble methods [18], [19] combine predictions

from multiple networks to assess model variance. Although effective, both approaches are computationally intensive and difficult to scale in multi-modal settings. As an alternative, evidential deep learning [20] directly estimates uncertainty by modeling evidence using Dirichlet or Gaussian distributions. However, most evidential approaches quantify uncertainty independently for each modality, overlooking cross-modal relationships. Ideally, modalities that are semantically aligned with the target should consistently yield lower uncertainty, while degraded or irrelevant modalities should exhibit higher uncertainty. Addressing the interdependence of uncertainties across modalities is therefore critical, yet remains underexplored in this field.

With the quantified uncertainty, we proceed to the multi-modal fusion and raise the second fundamental question: *How to handle conflicting predictions between modalities?* Traditional multi-modal fusion methods often assume that different modalities are semantically aligned with the target task [21], [22]. However, this assumption rarely holds in practice due to modality-specific noise, heterogeneity, or data incompleteness [23]. Recent methods such as Trusted Multi-view Classification (TMC) [24], [25] fuse predictions using Dempster's rule [26], [27], which integrates multiple evidences based on belief theory. While theoretically appealing, Dempster's rule may yield counter-intuitive results under strong conflicts. For instance, when two highly confident but contradictory modalities are combined, the resulting uncertainty may decrease paradoxically [28], [29], [30]. Additionally, the incremental nature of Dempster's rule can introduce bias: when a high-confidence prediction from one view contradicts with the previous results, the eventual prediction might be flipped disproportionately. In fact, the heterogeneity of multi-modal inputs also provides an opportunity to identify and mitigate biased outputs [31], thus conflict-aware fusion is essential.

In this paper, we propose a novel method for trustworthy multi-modal ultrasound fusion diagnosis, termed **TMUF**, which dynamically adapts the fusion process through *uncertainty calibration* and *conflict resolution*. TMUF is specifically designed to overcome the limitations of existing uncertainty-aware fusion approaches, providing a more robust and reliable integration of multi-modal ultrasound data. First, we introduce a **cross-modal uncertainty calibration regularizer** to calibrate modality-wise uncertainty by enforcing consistency in relative uncertainty across modalities. Unlike prior evidence-based approaches that operate independently on unimodal inputs, our regularizer aligns each modality's uncertainty (or confidence) with the likelihood of its prediction being correct. This is achieved through normalized distribution matching across modalities. Once calibrated, more reliable modalities naturally obtain higher fusion weights, while less informative ones contribute proportionally less to the final decision.

Second, to handle prediction conflicts across modalities, we propose a **credibility-aware multi-modality fusion strategy**. This mechanism jointly considers cross-modal prediction consistency and modality-specific confidence to dynamically adjust fusion weights. In our formulation, *credibility* is defined as a modality's relative contribution to the final decision, determined by its confidence and consistency with other

modalities. Based on the quantified uncertainty, TMUF separates modalities into *credible* and *non-credible* groups. A reward-and-penalty mechanism is then applied: for credible modalities supported by others, higher confidence leads to larger fusion weights; for conflicting modalities, higher confidence is penalized to reduce their influence. This design ensures that the final prediction is predominantly determined by reliable modalities, while minimizing the impact of noisy or misleading inputs.

The main contributions of this work are summarized as follows:

- We propose a cross-modal uncertainty calibration regularizer that aligns modality-wise uncertainty with the correctness likelihood of unimodal predictions, enabling more reliable modalities to receive higher fusion weights.
- We introduce a credibility-aware multi-modality fusion strategy that evaluates both modality-specific confidence and inter-modality consistency, establishing a dynamic and conflict-aware fusion paradigm.
- We validate TMUF on two CEUS-based diagnostic tasks: breast lesion classification and liver cancer differentiation, achieving accuracies of 88.00% and 92.08%, respectively, demonstrating the effectiveness of dynamic uncertainty calibration and conflict resolution.

II. RELATED WORK

In this section, we review related studies on deep learning-based multi-modal fusion, focusing on two key aspects: *uncertainty modeling in deep learning* and *conflicting evidence combination*.

A. Uncertainty Modeling in Deep Learning

Uncertainty estimation is critical in machine learning, particularly in open clinical environments where predictions are affected by data noise, annotation variability, and model instability [31]. In multi-modal ultrasound, uncertainty is further amplified by modality-specific noise [32], [33], missing data, and inter-modal imbalance [34], [35].

Existing deep learning approaches to uncertainty modeling fall into three categories: Bayesian neural networks (BNNs), deep ensembles, and evidential deep learning (EvDL). BNNs [36] treat model parameters as probability distributions to enable probabilistic inference, but are computationally expensive and difficult to scale in multi-modal settings. Ensemble methods [18] improve reliability by aggregating predictions from multiple models, though at the cost of increased parameters and training time. EvDL [20] offers a lightweight alternative by modeling uncertainty through belief masses under the Dempster-Shafer theory (DST) [26], [27]. DST extends Bayesian reasoning via subjective logic, enabling belief fusion across modalities using Dempster's rule.

B. Conflicting Evidence Combination

In real-world multi-modal learning, inputs often exhibit semantic misalignment or contradiction [37], [38]. For instance, BUS and CEUS may suggest HCC, while parametric

US indicates ICC features (Fig. 1), posing challenges for consistent and trustworthy fusion.

Early fusion strategies rely on heuristics [39], [40], [41], assigning fixed priorities to modalities, which lack adaptability in complex cases. Later works employ multi-view outlier detection [42], [43], [44] to identify unreliable modalities via affinity comparisons. More recently, uncertainty-aware fusion methods [45], [46], [47], [48] dynamically down-weight uncertain modalities. Trusted multi-view classification (TMC) [24] follows this direction, fusing evidence using Dempster's rule. However, in high-confidence but conflicting cases, TMC may paradoxically reduce uncertainty and produce unreliable outputs [22]. To mitigate such issues, Smarandache and Dezert [49] proposed proportional conflict redistribution, while Yager [50] suggested assigning conflicting mass to the universal set, acknowledging ambiguity. Both approaches align with key fusion principles: convergence, conflict-driven uncertainty, and consistency.

III. METHOD

This section details the proposed Trustworthy Multi-modal Ultrasound Fusion (TMUF) framework. The overall methodology includes: 1) formal problem definition (Sec. III-A), 2) evidence-based uncertainty estimation (Sec. III-B), 3) cross-modal calibration (Sec. III-C), 4) credibility-aware fusion strategy (Sec. III-D), 5) hybrid loss function design (Sec. III-E), and 6) implementation details (Sec. III-F).

A. Architecture

An overview of the proposed Trustworthy Multi-modal Ultrasound Fusion (TMUF) framework is illustrated in Fig. 2. Given a multi-modal dataset $\mathbb{D} = \{(x_i^{bus}, x_i^{ceus}, x_i^{elast}, \dots, x_i^{tic}), y_i\}_{i=1}^N$, where y_i is the class label and x_i^{bus} , x_i^{ceus} , x_i^{elast} , x_i^{tic} denote the B-mode, contrast-enhanced, and elastography ultrasound, time-intensity curves respectively, our goal is to dynamically integrate multiple modalities for reliable lesion classification. The core challenges lie in: modeling modality-wise uncertainty and resolving cross-modal conflicts.

To this end, we formulate the fusion pipeline in four steps: 1) uncertainty u^m is estimated for each modality m , and is constrained to be proportional to the classification error $d(\mathbf{p}^m, \mathbf{p}^*)$; 2) a cross-modal uncertainty calibration regularizer aligns the prediction error vector $\mathbf{d}$ with the uncertainty vector $\mathbf{u}$; 3) a discrepancy matrix $\mathbf{S} \in \mathbb{R}^{M \times M}$ is computed to quantify semantic alignment between modalities, which is further recalibrated using uncertainty-aware scale factors w_{mn} ; 4) a credibility-aware fusion scheme computes adaptive weights a_m using a reward-penalty mechanism, enabling evidence fusion guided by both uncertainty and inter-modal consistency.

Notably, TMUF is implemented as a generalizable multi-modal fusion framework with a modular design, enabling it to handle varying types and numbers of modalities. Each modality is processed independently, and its uncertainty calibration and credibility-adaptive fusion operate at the evidence level rather than relying on fixed modality identities. When certain modalities are missing, TMUF dynamically reallocates fusion weights based on the estimated reliability and consistency.

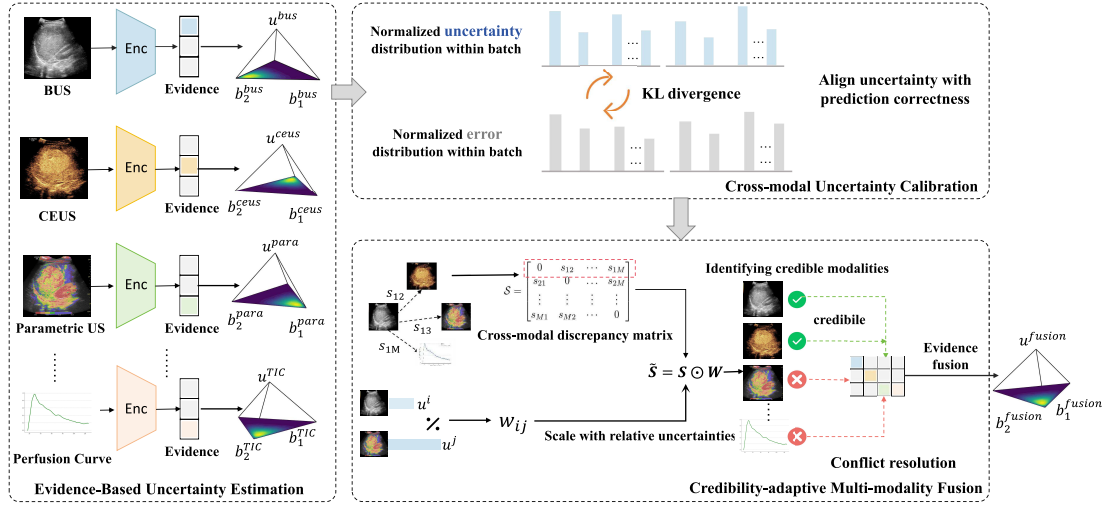


Fig. 2. The framework of the proposed trustworthy multi-modal ultrasound fusion method (TMUF). As illustrated, TMUF first estimates modality-wise prediction confidence by aligning prediction correctness with evidence-based uncertainty through cross-modal uncertainty calibration mechanism. Based on that, credibility-aware multi-modality fusion module is employed to assess the reliability of each modality, enabling to reward credible views and penalize conflicting or uncertain ones during dynamic fusion.

B. Evidence-Based Uncertainty Estimation

To assess the trustworthiness of each modality under noisy or biased conditions, we adopt evidence-based uncertainty (EvU) estimation grounded in Dempster–Shafer Theory (DST). In classification, this is implemented using subjective logic, which maps the parameters of a Dirichlet distribution to belief and uncertainty.

Given a K -class classification task, the Dirichlet distribution over the class probability vector $\mathbf{p} = [p_1, \dots, p_K]$ is defined as:

$$D(\mathbf{p}|\alpha) = \begin{cases} \frac{1}{B(\alpha)} \prod_{i=1}^K p_i^{\alpha_i-1} & \text{if } \mathbf{p} \in \mathcal{H}_K \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Here, α denotes the concentration parameters, with $\alpha_i = e_i + 1$, where e_i is the evidence assigned to class i . $B(\cdot)$ is the multinomial beta function and $\mathcal{H}_K$ is the K -dimensional probability simplex.

The belief mass b_i and total uncertainty u are computed as:

$$u = \frac{K}{S}, \quad b_i = \frac{e_i}{S}, \quad \text{where } S = \sum_{i=1}^K \alpha_i \quad (2)$$

The mean class probability is $\hat{p}_i = \frac{\alpha_i}{S}$. Notably, the total belief and uncertainty satisfy $\sum_{i=1}^K b_i + u = 1$. Higher uncertainty indicates weaker evidence and potentially incorrect predictions, motivating the need for further calibration.

C. Cross-Modal Uncertainty Calibration

Most existing EvU methods compute uncertainty per modality independently, ignoring inter-modal dependencies. We argue that uncertainty values should reflect prediction performance, and propose a cross-modal calibration scheme based on ranking consistency.

For multi-modal US fusion diagnosis, we first calculate the likelihood of prediction correctness of each modality. In this work, cross-entropy is used to measure the distance

between the predicted class probability $\mathbf{p}^m$ and the true label $\mathbf{p}^*$. Then, we constrain the modality-wise uncertainty u^m to be proportional to the distance $d(\mathbf{p}^m, \mathbf{p}^*)$:

$$u^m = \lambda d(\mathbf{p}^m, \mathbf{p}^*) = -\lambda \sum_i p_i^* \log(p_i^m) \quad (3)$$

where λ is an implicit proportionality constant.

Over a batch of size $\mathcal{B}$, we compute the $M \times \mathcal{B}$ -dimensional **distance vector** $\mathbf{d}$ and the **uncertainty vector** $\mathbf{u}$:

$$\mathbf{d} = [d_{bus}^1, \dots, d_{bus}^M, d_{ceus}^1, \dots, d_{ceus}^M, \dots, d_{tic}^1, \dots, d_{tic}^M] \quad (4)$$

$$\mathbf{u} = [u_{bus}^1, \dots, u_{bus}^M, u_{ceus}^1, \dots, u_{ceus}^M, \dots, u_{tic}^1, \dots, u_{tic}^M] \quad (5)$$

These are normalized into distributions $\tilde{\mathbf{d}}$ and $\tilde{\mathbf{u}}$, and aligned via Jensen-Shannon divergence:

$$\mathbf{v} = \frac{1}{2}(\tilde{\mathbf{d}} + \tilde{\mathbf{u}}) \quad (6)$$

$$\mathcal{L}_{cal} = \frac{1}{2} \text{KL}(\tilde{\mathbf{d}}\|\mathbf{v}) + \frac{1}{2} \text{KL}(\tilde{\mathbf{u}}\|\mathbf{v}) \quad (7)$$

Minimizing $\mathcal{L}_{cal}$ imposes a cross-modal uncertainty calibration regularizer, which encourages the uncertainty values to faithfully reflect prediction capacity, thereby enhancing reliability for downstream fusion. Notably, although $\mathcal{L}_{cal}$ leverages cross-entropy to measure predictive performance, it is not optimized as a direct loss function. In our implementation, the term $d(\mathbf{p}^m, \mathbf{p}^*)$ is detached from the computation graph and does not contribute gradients.

D. Credibility-Aware Multi-Modality Fusion

Given the intrinsic heterogeneity and randomness of multi-modal ultrasound data, the semantic alignment between modalities cannot be guaranteed. To mitigate the risk of misleading predictions due to modality conflicts, we propose a credibility-aware fusion strategy that jointly considers inter-modal consistency and intra-modal uncertainty. Rather than performing pairwise fusion or relying on fixed rules, our

method dynamically evaluates each modality's credibility and adjusts its contribution to the final prediction accordingly.

1) *Identifying Credible Modalities*: Specifically, this strategy first splits all modalities into credible and non-credible ones based on inter-modal alignment and intra-modal uncertainty. We begin by quantifying the semantic alignment among modalities. For each modality m , we compute its discrepancy with modality n as:

$$s_{mn} = \sqrt{\frac{1}{2} (\hat{\mathbf{p}}_m - \hat{\mathbf{p}}_n)^\top (\hat{\mathbf{p}}_m - \hat{\mathbf{p}}_n)} \quad (8)$$

where $\hat{\mathbf{p}}_m$ is the mean class probability in Section III-B. This yields a symmetric discrepancy matrix $\mathbf{S} \in \mathbb{R}^{M \times M}$, where M is the number of modalities. The average discrepancy for modality m is:

$$s_m = \frac{1}{M-1} \sum_{n \neq m} s_{mn} \quad (9)$$

where s_m measures the degree of semantic alignment between m_{th} modality with other. Apparently, a lower s_m indicates more support from the remaining views, while a higher value suggests a potentially conflicting or outlier view.

Using the mean discrepancy $\theta = \frac{1}{M} \sum_m s_m$ as a threshold, modalities are categorized as:

$$\mathcal{M}_{\text{credible}} = \{\mathcal{M}_m \mid s_m \leq \theta\}, \quad (10)$$

$$\mathcal{M}_{\text{non-credible}} = \{\mathcal{M}_m \mid s_m > \theta\} \quad (11)$$

To further refine this process, we introduce a scale factor w_{mn} that down-weights contributions from less reliable modalities:

$$w_{mn} = \begin{cases} \frac{u_m}{u_n}, & \text{if } u_m < u_n \\ 1, & \text{otherwise} \end{cases} \quad (12)$$

The recalibrated discrepancy matrix is obtained by element-wise multiplication:

$$\tilde{\mathbf{S}} = \mathbf{S} \odot \mathcal{W} \quad (13)$$

Accordingly, the updated credibility threshold becomes $\tilde{\theta} = \frac{1}{M} \sum_m \tilde{s}_m$, which gives more weight to comparisons among reliable modalities, particularly useful to reduce the influence of unreliable views when conflict arises.

2) *Dynamic Fusion With Credibility Grouping*: Once modalities are divided into credible and non-credible groups, we assign weights using credibility-aware reweighting functions:

$$a_m = e^{-\bar{u}_m}, \quad \text{if } m \in \mathcal{M}_{\text{credible}} \quad (14)$$

$$a_m = e^{-(1+\bar{u}_{\max}-\bar{u}_m)}, \quad \text{if } m \in \mathcal{M}_{\text{non-credible}} \quad (15)$$

where $\bar{u}_m$ represents the normalized uncertainty, $\bar{u}_m = u_m / \sum_m u_m$. $\bar{u}_{\max}$ is the maximum normalized uncertainty. This formulation ensures that, for credible modalities, lower uncertainty leads to higher weight. For non-credible ones, we consider the contribution of these low-confidence modalities under two typical scenarios:

- **Low-confidence but aligned prediction**: If a low-confidence modality m happens to align with others (i.e., relatively small s_{mn}), it may still be classified as credible. However, due to its high uncertainty u_m , its final fusion weight is exponentially down-weighted

as $\exp(-u_m)$ (Eq. 14), typically close to $\exp(-1)$. This prevents it from dominating the fusion while preserving potentially useful signals.

- **Low-confidence and conflicting prediction**: If a low-confidence modality m diverges from others, the resulting discrepancies s_{mn} and its average s_m increase substantially, likely surpassing the threshold θ , causing it to be marked non-credible and penalized (Eq. 15). In this case, its weight also converges toward $\exp(-1)$, serving as a safeguard against severe misguidance.

Based on that, the smoothed weight $\bar{a}_m$ is obtained using a temperature-softmax:

$$\bar{a}_m = \frac{e^{a_m/\eta}}{\sum_{m=1}^M e^{a_m/\eta}} \quad (16)$$

where η is a temperature factor that adjusts sensitivity to correlation variance. Finally, we perform dynamic modality fusion at the evidence level:

$$\mathbf{e}_f = \sum_m \bar{a}_m \mathbf{e}_m \quad (17)$$

Based on $\mathbf{e}_f$ ¹, we can derive the class belief and the overall uncertainty. In comparison with TMC-like method, our TMUF method addresses those limitations mentioned in Section I by:

- Separating modalities into credibility groups before fusion, preventing unreliable views from dominating the decision;
- Penalizing high-confidence yet inconsistent modalities through the credibility-aware reweighting function;
- Applying a temperature-controlled softmax to smooth fusion weights, reducing sensitivity to extreme confidence disparities.

E. Loss Function

To effectively train the TMUF framework, we propose a hybrid loss that integrates classification supervision $\mathcal{L}_{cls}$, uncertainty calibration $\mathcal{L}_{cal}$, and prior regularization $\mathcal{L}_{kl}$. The overall loss $\mathcal{L}_B$ is formulated as:

$$\mathcal{L}_B = \frac{1}{n} \sum_i [\mathcal{L}_{cls}(\alpha_i) + \lambda_k \mathcal{L}_{kl}(\alpha_i)] + \lambda_c \mathcal{L}_{cal} \quad (18)$$

where $\alpha_i = \mathbf{e}_{i,f} + 1$ denotes the Dirichlet parameters derived from the fused evidence $\mathbf{e}_{i,f}$ of the i -th subject. The terms λ_k and λ_c are annealing weights for regularization and calibration losses, respectively. During training, we gradually increase these weights to balance learning stability and uncertainty alignment. Each component plays a distinct role:

1) *Supervised Classification Loss $\mathcal{L}_{cls}$* : This term ensures that the final fused prediction aligns with the ground truth labels. It is computed as the expected cross-entropy under the predictive Dirichlet distribution:

$$\mathcal{L}_{cls}(\alpha_i) = \int \left[\sum_{j=1}^K -y_{ij} \log p_{ij} \right] \frac{1}{B(\alpha_i)} \prod_{j=1}^K p_{ij}^{\alpha_{ij}-1} dp_i$$

¹For simplicity, we omit the subscript i for subject index.

$$= \sum_{j=1}^K y_{ij} (\phi(S_i) - \phi(\alpha_{ij})) \quad (19)$$

where $S_i = \sum_j \alpha_{ij}$, and $\phi(\cdot)$ is the digamma function. Notably, this loss is only computed from the fused evidence $\mathbf{e}_{i,f}$, without optimizing individual modalities, which avoids overfitting to modality-specific predictions and preserves inter-modal diversity.

2) *Prior Regularization $\mathcal{L}_{kl}$* : To penalize overconfident predictions, especially for incorrect classes, we introduce a KL divergence loss between the adjusted Dirichlet $\tilde{\alpha}_i$ and a uniform prior::

$$\tilde{\alpha}_i = y_i + (1 - y_i) \odot \alpha_i \quad (20)$$

$$= \log \left(\frac{\Gamma(\tilde{S}_i)}{\Gamma(K) \prod_{j=1}^K \Gamma(\tilde{\alpha}_{ij})} \right) + \sum_j (\tilde{\alpha}_{ij} - 1) [\phi(\tilde{\alpha}_{ij}) - \phi(\tilde{S}_i)] \quad (21)$$

This regularization encourages low evidence for non-target classes and curbs false certainty.

3) *Uncertainty Calibration Loss $\mathcal{L}_{cal}$* : As defined in Section III-C, this term enforces that each modality's uncertainty positively correlates with its prediction error. Rather than minimizing uncertainty outright, this term aligns *how uncertain* a modality is with *how wrong* it tends to be, ensuring meaningful uncertainty estimates.

This loss formulation reflects two key principles: 1) Only the fused evidence is supervised for classification, avoiding the need to minimize individual modality losses. This discourages overconfident memorization and promotes diversity. 2) Modality-level uncertainty is calibrated and regularized indirectly through $\mathcal{L}_{cal}$ and $\mathcal{L}_{kl}$, which together ensure that unreliable modalities are appropriately down-weighted in fusion (Section III-D).

F. Implementation Details

The TMUF framework is implemented in PyTorch and all experiments are conducted on a single NVIDIA RTX 3090 GPU. TMUF is trained using the Adam optimizer with an initial learning rate of 1×10^{-3} . Training is performed for 30 epochs with a batch size of 32, and early stopping is applied based on validation accuracy. A step decay schedule reduces the learning rate starting from the 15th epoch. All modality-specific encoders are initialized with CLIP-pretrained Vision Transformer (ViT) backbones and fine-tuned during training. The fusion softmax temperature η is set to 0.8 by default. The loss weights λ_k and λ_c , corresponding to the KL divergence and uncertainty calibration terms, are initialized at 0.2 and linearly increased to 1.0 throughout training.

IV. EXPERIMENTS

In this section, we first compare our TMUF method with several state-of-the-art methods. Then, we validate the effectiveness of cross-modal uncertainty calibration and credibility-aware multi-modality fusion of this method. After that, we further perform ablation experiments by removing the

core components. Finally, we analyze the sensitivity of various modality combinations and provide visualization results of modality contributions.

A. Datasets and Metrics

We evaluate the proposed TMUF framework on two multi-modal ultrasound datasets: a public breast cancer dataset (TDF-Net) [51] and a private liver cancer dataset (TMUF). For a fair evaluation, each dataset is randomly partitioned into training (60%), validation (20%), and test (20%) sets, while preserving class ratio. To reduce the impact of randomness, the splitting process is repeated five times, and we report the average and standard deviation of performance over five independent runs. Diagnostic performance is assessed using standard classification metrics, including accuracy (ACC), area under the receiver operating characteristic curve (AUC), sensitivity (SEN), specificity (SPE), precision, and F1-score. Hepatocellular carcinoma (HCC) and malignant breast lesions are treated as the positive class.

1) *Breast Dataset*: The public breast dataset contains 248 breast lesion cases, comprising 145 benign and 103 malignant samples. As mentioned in [51], all patients underwent core needle biopsy or surgical resection following ultrasound-based screening. Each case includes multi-modal ultrasound images acquired from three modalities, enabling fine-grained fusion across structural, functional, and mechanical dimensions of breast tissue.

- **B-mode Ultrasound (BUS)**: Three views per patient to capture different anatomical planes, including longitudinal, transverse, and radial orientations. Each view is treated as an independent modality.
- **Doppler Ultrasound (DUS)**: One view per patient for assessing vascular characteristics.
- **Elastography Ultrasound (EUS)**: Two views per patient to measure tissue stiffness.

2) *Liver Dataset*: The private liver dataset includes 238 cases from Nanjing Drum Tower Hospital—136 hepatocellular carcinoma (HCC) and 102 intrahepatic cholangiocarcinoma (ICC). All 238 cases were pathologically diagnosed through biopsy or surgical specimen analysis. Diagnoses were established by experienced pathologists and used as the gold-standard labels for distinguishing HCC and ICC. Ultrasound imaging was performed using a Philips iU22 scanner, with contrast-enhanced ultrasound (CEUS) conducted using second-generation SonoVue agents (Bracco SpA, Milan, Italy) at a low mechanical index. Each dual-view CEUS video spans about 3 minutes. Data collection was conducted with informed consent from all participants and approved by the Ethics Review Committee of Nanjing Drum Tower Hospital. Each case includes five complementary modalities, enabling comprehensive characterization of liver lesions across spatial, functional, and temporal dimensions:

- **B-mode Ultrasound (BUS)**: Gray-scale images acquired from the optimal scanning plane
- **Contrast-Enhanced Ultrasound (CEUS)**: CEUS snapshots taken at the peak perfusion phase to reflect microvascular patterns.

- **Parametric Ultrasound (PARA):** A CEUS-derived modality in which each pixel encodes the arrival time of contrast agents, reflects spatial perfusion timing. PARA is automatically computed by the Philips iU22 system as a pseudo-color parametric map.
- **Morphological Features:** Radiomics-based features representing lesion shape, boundary, and internal texture characteristics.
- **Perfusion Time–Intensity Curve (TIC):** Quantifies dynamic contrast enhancement over time, reflecting perfusion kinetics.

Notably, directly processing full CEUS video sequences would introduce substantial computational overhead and memory usage due to high frame rates. Instead, our framework captures temporal dynamics through CEUS snapshot, parametric ultrasound, and TIC, retaining key perfusion dynamics while ensuring efficiency.

B. Competing Methods

To evaluate the effectiveness of the proposed TMUF framework, we compare it against several state-of-the-art multi-modal fusion methods, encompassing both classical and uncertainty-aware approaches:

- **DCCA [53]:** Deep Canonical Correlation Analysis extends traditional linear CCA by learning hierarchical nonlinear transformations that maximize correlation across multiple views.
- **DCCAE [54]:** Deep Canonically Correlated Auto-Encoders combine CCA with autoencoders to simultaneously maximize inter-view correlation and reconstruct modality-specific features through a dual-stream structure.
- **MVCNN [55]:** Multi-view Convolutional Neural Network aggregates localized view-specific features into a shared global representation using parallel convolutional encoding and cross-view fusion.
- **MM Dynamics [56]:** A dynamic fusion framework that evaluates modality quality hierarchically, integrating both feature granularity and modality completeness to allocate adaptive weights based on estimated reliability.
- **TMC [24]:** Trusted Multi-view Classification models uncertainty using Dirichlet distributions under evidence theory, and employs Dempster’s rule for credibility-aware fusion.²
- **QMF [57]:** Quality-aware Multi-modal Fusion leverages real-time quality assessment and attention-based recalibration to suppress noisy modalities, enhancing robustness in low-quality scenarios.³
- **RCML [22]:** Reliable Conflictive Multi-view Learning explicitly models conflict and certainty between modalities. It estimates both shared and view-specific reliabilities to guide robust decision fusion under severe modality disagreement.⁴

- **TDF-Net [51]:** Trusted Dynamic Feature Fusion Network enforces semantic consistency across modalities via contrastive clustering loss. A Dirichlet-based dynamic fusion module quantifies per-modality reliability and adaptively integrates features based on uncertainty.⁵
- **BMU-Net [58]:** Breast Multimodal Ultrasound Network employs parallel branches to extract modality-specific features from mammography and ultrasound (B-mode, elastography, and Doppler ultrasound). A multi-level cross-attention fusion module integrates complementary features hierarchically to improve breast lesion classification performance.⁶

To ensure fair comparison, all methods were trained and evaluated with consistent input modalities across datasets. For breast lesion classification, BUS, EUS, and DUS were used for all models; for liver cancer differentiation, the full set of available modalities (BUS, CEUS, PARA, Morphology, and TIC) was employed. When necessary, baseline models were extended with modality-specific projection heads to accommodate non-image inputs. For TIC, the raw time–intensity curve was treated as a one-dimensional temporal sequence and processed using a 1D convolutional network. For Morphological features, 2D morphological descriptors were extracted from lesion regions using the PyRadiomics toolkit, following standardized radiomics feature definitions. Both branches were trained jointly with imaging modalities in an end-to-end manner.

C. Comparison on Diagnostic Performance

We evaluate TMUF against several state-of-the-art multi-modal fusion methods on two diagnostic tasks: breast lesion classification and primary liver cancer differentiation. Quantitative results are summarized in Table I.

Experimental results reveal that our TMUF method consistently demonstrates significant performance improvements on both diagnostic tasks. For breast lesion classification, TMUF achieves the highest overall performance across all metrics. Notably, it obtains an AUC of 95.44%, substantially outperforming other methods, whose AUCs remain below 94%. The baseline DCCA shows the weakest performance (AUC: 87.13%), while BMU-Net is the strongest competitor (AUC: 93.83%) but still lags behind TMUF. In terms of sensitivity, which is critical for detecting malignant cases, most methods perform suboptimally (range: 70.48%–82.86%). TMUF distinctly outperforms with a SEN of 83.81%, while also achieving the best specificity (91.03%) and accuracy (88.00%). This indicates TMUF’s balanced capability in identifying both positive and negative cases, reducing the risk of missed diagnoses in malignant conditions. We also benchmarked TMUF against the domain-specific solution TDF-Net under identical experimental settings. TMUF achieved consistently higher performance across multiple metrics, with improvements in AUC (95.44% vs. 92.55%), ACC (88.00% vs. 82.40%), and F1-score (85.46% vs. 70.48%). These gains can be attributed

²<https://github.com/hanmenghan/TMC>

³<https://github.com/QingyangZhang/QMF>

⁴<https://github.com/jiajunsj/RCML>

⁵https://github.com/TimesXY/TDF_Net

⁶<https://github.com/Qian-IMMULab/BMU-Net>

TABLE I
DIAGNOSTIC PERFORMANCES COMPARISON OF VARIOUS METHODS. THE TERMS A AND B IN “A±B” DENOTE THE MEAN AND STANDARD DEVIATION (STD)

Method	AUC	ACC	SEN	SPE	F1-score	Precision
Differentiation of Benign and Malignant Breast Lesion						
DCCA	87.13±0.25	80.40±4.77	76.19±5.83	83.45±7.48	76.63±5.29	77.57±8.49
DCCAE	87.49±0.79	79.20±2.28	70.48±6.21	85.52±4.50	73.93±3.06	78.24±4.51
MVCNN	88.08±3.90	80.00±3.74	70.48±7.82	86.90±3.78	74.62±5.31	79.72±5.22
MM Dynamics	91.99±1.37	84.80±1.79	79.05±7.97	88.97±5.11	81.25±2.91	84.49±5.72
TMC	90.25±1.92	81.20±3.03	79.05±8.65	82.76±8.79	77.90±3.17	77.98±7.94
QMF	90.67±1.28	84.40±4.34	80.00±8.52	87.59±8.99	81.18±4.92	83.44±8.62
RCML	91.46±2.23	84.00±3.16	80.95±3.37	86.21±3.45	80.97±3.63	81.03±4.34
TDF-Net	92.55±0.41	82.40±2.19	76.89±4.03	91.03±6.26	70.48±9.16	86.17±7.32
BMU-Net	93.83±1.68	86.00±3.74	82.86±9.96	88.28±3.08	82.97±5.73	83.74±2.55
TMUF (Ours)	95.44±0.90	88.00±2.45	83.81±7.22	91.03±7.15	85.46±2.71	88.19±8.55
Differentiation of Primary Liver Cancer						
DCCA	80.87±1.35	74.58±3.09	59.14±8.70	86.06±4.98	66.37±5.72	76.76±4.83
DCCAE	85.18±1.68	75.83±2.38	63.05±9.18	85.42±7.72	68.88±4.68	77.69±8.84
MVCNN	85.86±4.33	77.92±1.86	59.71±9.62	91.96±7.92	69.61±5.56	87.48±9.64
MM Dynamics	88.28±2.96	81.25±3.29	71.48±5.83	88.28±8.14	76.50±2.76	83.28±8.47
TMC	90.14±1.51	82.92±4.75	71.90±4.93	91.24±6.71	78.42±5.22	86.73±9.43
QMF	91.04±2.36	85.00±3.42	75.24±3.98	92.59±6.93	81.53±3.33	89.66±8.74
RCML	93.13±0.85	84.58±2.80	72.19±8.60	93.52±6.40	79.61±4.29	90.37±9.46
Guo et al. [52]	90.00±2.82	84.58±3.49	78.24±4.30	89.18±5.15	81.09±3.84	84.45±6.52
TMUF (Ours)	95.90±1.33	92.08±2.72	86.57±6.14	96.30±4.54	90.46±3.30	95.22±5.58

to TMUF’s credibility-aware fusion, which aligns modality-specific uncertainty with predictive reliability and adaptively down-weights less informative modalities. In contrast, TDF-Net’s feature-level coherence strategy may be less robust when certain modalities contain substantial noise or artifacts.

In liver cancer classification, TMUF again leads across all metrics, achieving a top AUC of 95.90% and an ACC of 92.08%, well above the best baseline RCML (AUC: 93.13%, ACC: 84.58%). Its SEN of 86.57% and SPE of 96.30% demonstrate superior performance in both lesion detection and avoidance of false positives. We further adapted the approach of Guo et al. [52] to our private liver dataset (BUS, CEUS, PARA, Morphology, TIC), enabling multi-modal classification of HCC and ICC. This method underperforms with an average AUC of 90% and ACC of 84.58%, significantly lower than TMUF (95.90% AUC, 92.08% ACC). Compared to breast lesion analysis, competing methods on liver cancer also exhibit skewed performance—high SPE but noticeably lower SEN. In contrast, TMUF maintains a consistently high performance across all evaluation dimensions.

Overall, TMUF shows strong generalization (higher SEN without sacrificing SPE) across both breast and liver cancer datasets. Compared to static fusion models (e.g., DCCA, MVCNN) and uncertainty-aware baselines (e.g., TMC, QMF), TMUF’s credibility-adaptive mechanism and calibrated uncertainty fusion yield consistently higher diagnostic accuracy.

D. Comparison of Uncertainty Estimation

As introduced in Section III-C, TMUF incorporates a cross-modal uncertainty calibration framework to enhance fusion robustness by aligning prediction error with modality-wise uncertainty. To assess its effectiveness, we compare TMUF against four widely-used uncertainty modeling techniques on the two datasets: 1) **Temperature Scaling (Temp)** [59];

2) **Probabilistic Models (Prob)** [60]; 3) **Dropout** [61]; 4) **Deep Ensembles** [18]. All comparative methods were integrated into a unified backbone with identical encoders, fusion strategy, and training protocols, ensuring fair and controlled evaluation. Experimental results are reported in Fig. 3.

As illustrated, several key observations can be made: 1) TMUF achieves the best performance in multi-modal ultrasound fusion, with an AUC of 95.44% and ACC of 88.00%,

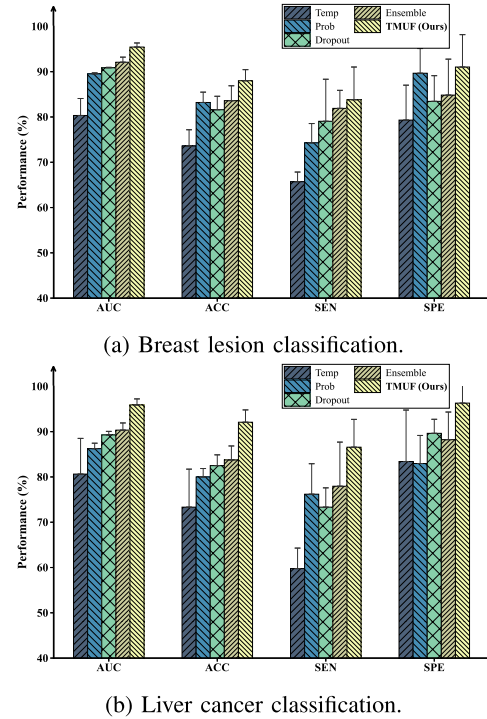


Fig. 3. Comparison of different uncertainty estimation methods across two datasets.

significantly outperforming Temperature Scaling (80.33%) and Probabilistic Models (89.53%) on the breast dataset. This gain results from TMUF's dynamic uncertainty calibration and conflict-aware fusion, which adaptively weight reliable modalities while suppressing noisy ones. 2) Temperature Scaling, based on static calibration, performs the worst. Its inability to account for modality-dependent uncertainty limits its applicability in complex diagnostic settings, highlighting the limitations of static approaches under dynamic conditions. 3) Although Dropout and Deep Ensembles introduce stochasticity or model diversity, their performance remains suboptimal (AUCs: 90.87% and 92.09%).

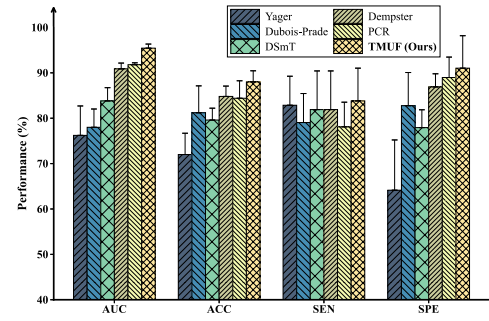
On the liver dataset, TMUF again achieves the highest overall performance, with notable gains in sensitivity (86.57%) and specificity (96.30%), both critical for reliable HCC–ICC differentiation. Competing methods such as Dropout and Deep Ensembles achieve comparable AUCs (approximately 90%) but exhibit reduced sensitivity (75%), similar to probabilistic models. These methods lack explicit modeling of inter-modal uncertainty dependencies—an essential factor addressed by TMUF for reliable evidence fusion.

E. Comparison of Evidence Fusion

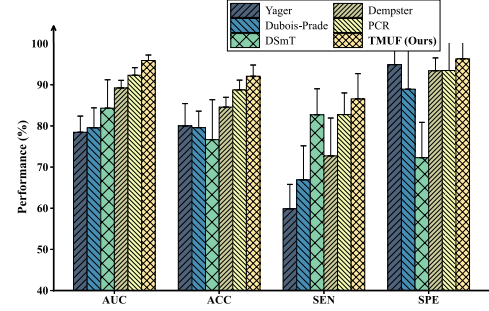
As introduced in Section III-D, TMUF resolves multi-modal prediction conflicts by dynamically evaluating cross-modal consistency and modality-specific uncertainty. To assess its effectiveness, we compare TMUF with five established conflict resolution methods on the two datasets: 1) **Dubois–Prade rule** [62]; 2) **Yager's rule** [50]; 3) **Dezert–Smarandache Theory (DSmT)** [63]; 4) **Dempster's rule** [64]; 5) **Proportional Conflict Redistribution (PCR)** [49]. To ensure a fair evaluation, we hold all other factors constant, including training loss, network capacity, and modality configuration.

From Fig. 4, several key observations emerge: 1) TMUF achieves the best performance on the breast dataset, with an AUC of 95.44% and ACC of 88.00%, substantially surpassing Yager (AUC: 76.22%) and Dubois–Prade (AUC: 78.00%). This gain stems from TMUF's ability to assign higher fusion weights to consistent and reliable modalities, whereas Yager and Dubois–Prade rely on static fusion rules that cannot adapt to modality variation. 2) DSmT and Dempster offer improved handling of evidence conflict through rule-based aggregation, achieving moderate AUCs of 83.81% and 90.87%, respectively. Among the baselines, PCR performs best (AUC: 91.79%) due to its proportional redistribution of conflict mass. However, it still falls short of TMUF, which integrates uncertainty calibration to dynamically adapt to changing modality reliability.

For the liver cancer dataset, TMUF continues to achieve the highest performance, with an AUC of 95.90% and ACC of 92.08%, outperforming other competing fusion strategies. Notably, although Dempster's rule yields a relatively strong AUC (around 90%), its sensitivity remains lower than that of TMUF (73% vs. 86.57%). Other static rules such as Yager and Dubois–Prade show marked performance degradation, likely due to their limited capacity to handle modality conflicts under complex semantic variations. Overall, TMUF's performance



(a) Breast lesion classification.



(b) Liver cancer classification.

Fig. 4. Comparison of different evidence fusion strategies across two datasets.

TABLE II
COMPARISON OF VARIOUS MODALITY COMBINATIONS

Modality	AUC	ACC	SEN	SPE	F1-score
BUS	79.77±2.30	76.40±4.77	70.48±7.06	80.69±9.32	71.56±4.65
DUS	81.74±5.87	74.80±4.15	61.90±6.73	84.14±6.26	67.32±5.40
EUS	84.66±1.83	79.60±2.19	66.67±9.65	88.97±5.67	72.96±5.09
BUS+DUS	86.37±1.54	79.60±2.61	74.29±7.22	83.45±4.50	75.27±3.63
EUS+DUS	91.99±1.31	84.40±4.34	80.00±7.06	87.59±5.23	81.12±5.23
TMUF (All)	95.44±0.90	88.00±2.45	83.81±7.22	91.03±7.15	85.46±2.71

advantage highlights the importance of conflict resolution strategies that are not only aware of uncertainty but also responsive to inter-modal consistency under real-world clinical conditions.

F. Comparison of Multi-Modal Combination

Multi-modal fusion improves diagnostic accuracy by integrating complementary information across modalities. In medical ultrasound, combining structural, vascular, and stiffness-related features has proven particularly valuable. To assess the diagnostic contribution of different modality combinations, we benchmarked BUS, DUS, and EUS individually and in pairs on public breast dataset. Results are shown in Table II.

BUS alone achieves an AUC of 79.77%, reflecting its strength in capturing anatomical structures but limited sensitivity to vascular and elastic features. DUS performs slightly better (AUC: 81.74%), due to its ability to detect blood flow and microvascular changes. In contrast, EUS-alone achieves a single-modality AUC of 84.66%, underscoring the superior discriminative capability of elasticity features in breast lesion

TABLE III
COMPARISON OF VARIOUS VARIANTS. THE TERMS A AND B IN “A±B” DENOTE THE MEAN AND STANDARD DEVIATION (STD.). SYMBOLS: • USED, ○ UNUSED

Method	Components			Performance Metric				
	III-C	III-D.1	III-D.2	AUC	ACC	SEN	SPE	F1-score
V1	○	○	○	86.60±0.38	81.20±4.38	75.24±6.21	85.52±6.63	77.11±5.23
V2	•	○	○	90.31±1.76	84.40±2.61	80.00±6.21	87.59±8.30	81.24±1.99
V3	•	•	○	92.38±0.78	83.20±1.10	73.33±9.96	90.34±9.57	78.42±1.62
V4	•	○	•	90.15±0.85	79.20±7.56	68.57±8.91	86.90±2.89	72.86±9.22
V5	○	•	•	91.49±0.95	84.00±2.45	76.19±4.76	89.66±4.22	79.99±3.02
TMUF (Ours)	•	•	•	95.44±0.90	88.00±2.45	83.81±7.22	91.03±7.15	85.46±2.71

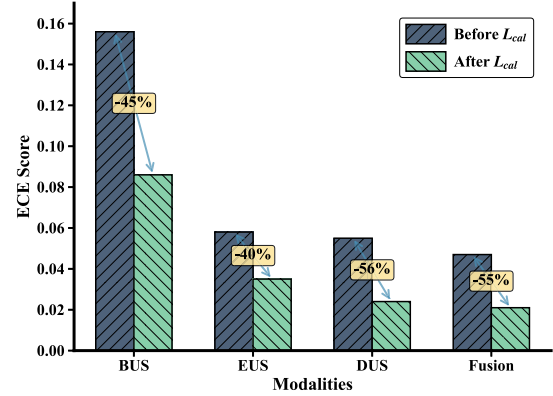
characterization. It should be noted that the single-modality AUCs reported in Table II (79.77%–84.66%) are lower than those reported in some previous studies [65], [66], which exceed 90%. This difference can be explained by: 1) our use of full-frame ultrasound images without manual ROI cropping; 2) dataset heterogeneity introduced by variations in acquisition protocols, scanner configurations, and patient populations; and 3) the adoption of a unified ViT-based backbone for all modalities, without modality-specific optimization to ensure fair and consistent evaluation.

Combining BUS and DUS yields a moderate improvement (AUC: 86.37%), suggesting partial complementarity. In contrast, the combination of EUS and DUS achieves a notably higher AUC of 91.99%, highlighting the synergistic benefits of integrating tissue stiffness and perfusion information. This combination outperforms BUS+DUS by a substantial margin, demonstrating that complementary physiological cues contribute more than structural similarity. Our TMUF method integrates all three modalities—BUS, DUS, and EUS—and achieves the best overall performance (AUC: 95.44%, ACC: 88.00%). This result confirms that effective multi-modal fusion, supported by uncertainty calibration and conflict-aware weighting, can fully leverage cross-modal information for enhanced diagnostic precision. TMUF’s ability to adaptively integrate diverse signals further underscores its potential in clinical ultrasound analysis.

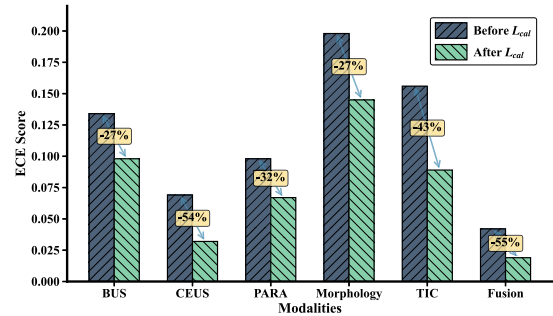
G. Ablation Study

To assess the contribution of each module, we conducted ablation experiments on public breast dataset by selectively removing the uncertainty calibration module (Section III-C) and the conflict resolution module (Section III-D). Table III summarizes the results for breast lesion classification.

Variant V1, with both modules removed, achieves the lowest performance (AUC: 86.60%, ACC: 81.20%), indicating the importance of both components. V2 retains only the calibration module, improving AUC to 90.31%, while V5 includes only the conflict resolution module, yielding a slightly higher AUC of 91.49%. However, both variants fall short of the full model. Additionally, V3 (*Discrepancy-only*) completely ignores non-credible modalities based on cross-modal discrepancy (Eqs. 12–13), and V4 (*Uncertainty-only*) assumes all modalities are credible and determines weights solely from uncertainty estimates (Eqs. 14–15). As shown in Table III, V3 underperforms TMUF by 3.06% AUC and 4.80% ACC,



(a) Breast lesion classification.



(b) Liver cancer classification.

Fig. 5. Calibration performance comparison measured by ECE scores. Lower ECE values mean better calibration.

indicating that complete exclusion of non-credible modalities discards partially correct information. V4 performs even worse due to the unfiltered inclusion of overconfident but incorrect modalities.

Our complete TMUF framework, combining both modules, achieves the best results (AUC: 95.44%, ACC: 88.00%). These findings confirm that neither module alone is sufficient to realize full performance, and that their joint optimization leads to a substantial performance gain. The synergy between uncertainty calibration and conflict-aware fusion is essential for robust multi-modal integration and improved diagnostic accuracy.

H. Calibration Performance Evaluation

To further quantify the impact of the calibration loss $\mathcal{L}_{cal}$, we evaluated its effect on both single-modality and fused

predictions using the expected calibration error (ECE), where lower values indicate better calibration and zero denotes perfect alignment between predicted confidence and empirical accuracy. The ECE is computed by partitioning predictions into M confidence bins, calculating the absolute difference between accuracy and average confidence within each bin, and taking the weighted average across bins. Experimental results on the breast and liver cancer datasets are presented in Figs. 5(a) and 5(b).

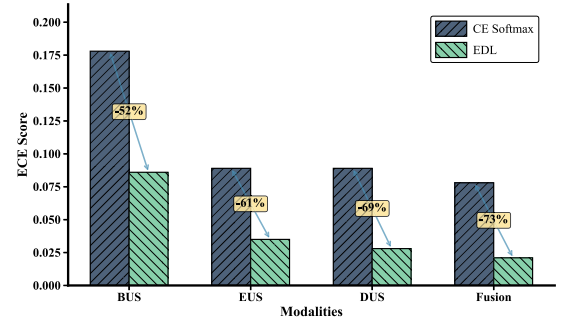
For the breast cancer dataset, incorporating $\mathcal{L}_{cal}$ consistently improved the calibration of all single modalities as well as the fused prediction. The BUS modality exhibited the largest relative ECE reduction (−45%), substantially alleviating its severe overconfidence when used alone. EUS and DUS achieved ECE reductions of −40% and −56%, respectively. At the fusion level, the ECE decreased by −55%, indicating that $\mathcal{L}_{cal}$ effectively promotes well-calibrated consensus predictions. Similar trends were observed on the liver cancer dataset, with BUS (−27%), CEUS (−54%), PARA (−32%), Morphology (−27%), and TIC (−43%) all showing substantial improvement. The fused output achieved the largest overall reduction of −55%, confirming that $\mathcal{L}_{cal}$ benefits both individual modality calibration and the final fusion output.

These results demonstrate that $\mathcal{L}_{cal}$ not only mitigates overconfidence in individual modalities but also enhances the calibration of the fused output. Although optimized at the fusion stage, $\mathcal{L}_{cal}$ indirectly improves per-modality calibration via the credibility-aware fusion mechanism, which down-weights modalities whose confidence is inconsistent with cross-modal consensus.

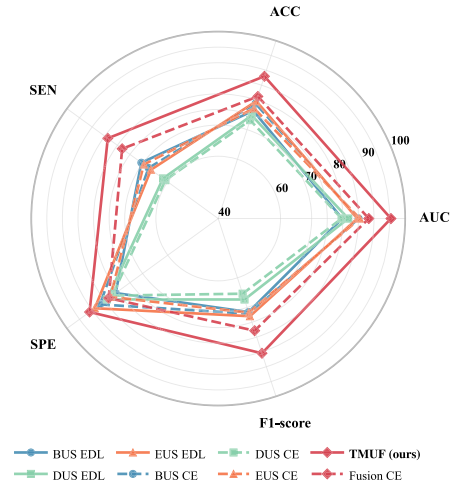
I. Comparison With Standard Softmax CE Training

To quantitatively evaluate the advantages of EDL within the TMUF framework, we compare it against a widely used CE fusion baseline in which the logits from modality-specific CE classifiers are averaged and then normalized via softmax. This ensemble-style logit averaging is a common approach to uncertainty modeling when explicit belief representations are unavailable. Both methods adopt identical network architectures, differing only in loss formulation and uncertainty representation.

As shown in Fig. 6(a), EDL consistently produces better-calibrated predictions than CE across all modalities, achieving relative ECE reductions of 52% (BUS), 61% (EUS), 69% (DUS), and 73% (Fusion). This improved calibration is critical for credibility-aware weighting, as modality contributions are directly modulated by their predicted uncertainty. Fig. 6(b) compares predictive metrics (AUC, ACC, SEN, SPE, F1-score) between CE and EDL variants. While CE occasionally surpasses EDL in isolated single-modality AUC (e.g., BUS and EUS), EDL delivers more balanced sensitivity–specificity trade-offs and superior fusion performance. Notably, TMUF (EDL-based) improves fusion AUC by +7.10% and ACC by +6.80% over CE-based fusion, with consistent gains across all other metrics. These results demonstrate that the structured uncertainty from EDL enables the fusion mechanism to more effectively suppress misleading modalities



(a) ECE score comparison between CE and EDL for each modality and fusion output. Lower ECE values mean better calibration.



(b) Radar chart comparing AUC, ACC, SEN, SPE, and F1-score across modalities and fusion results.

Fig. 6. Comparison between EDL-based training strategy and a standard softmax + cross-entropy (CE).

and emphasize consistent, credible ones. In summary, EDL provides well-calibrated, uncertainty-aware predictions that are essential for the proposed credibility-aware exponential fusion to reach its full potential. Although CE may yield marginal gains in isolated branches, it lacks the probabilistic structure necessary for robust multi-modal integration—precisely the area where TMUF achieves its decisive advantage.

V. DISCUSSION

A. Qualitative Analysis of Conflicting Modality Fusion

To examine the conflicting modality fusion in TMUF, we provide a detailed qualitative analysis under challenging multi-modal scenarios.

1) *Conflicting Yet Confident Modality Fusion:* Fig. 7(a-b) presents two representative scenarios in which different modalities produce high-confidence yet contradictory predictions. In the breast cancer case (Fig. 7a), all three BUS views misclassify the lesion as benign with moderate uncertainty scores (0.25–0.28), whereas the DUS and EUS branches predict malignancy, supported by Doppler flow patterns and elevated stiffness regions in the elasticity maps. In the liver cancer case (Fig. 7b), the BUS and Parametric US modalities predict

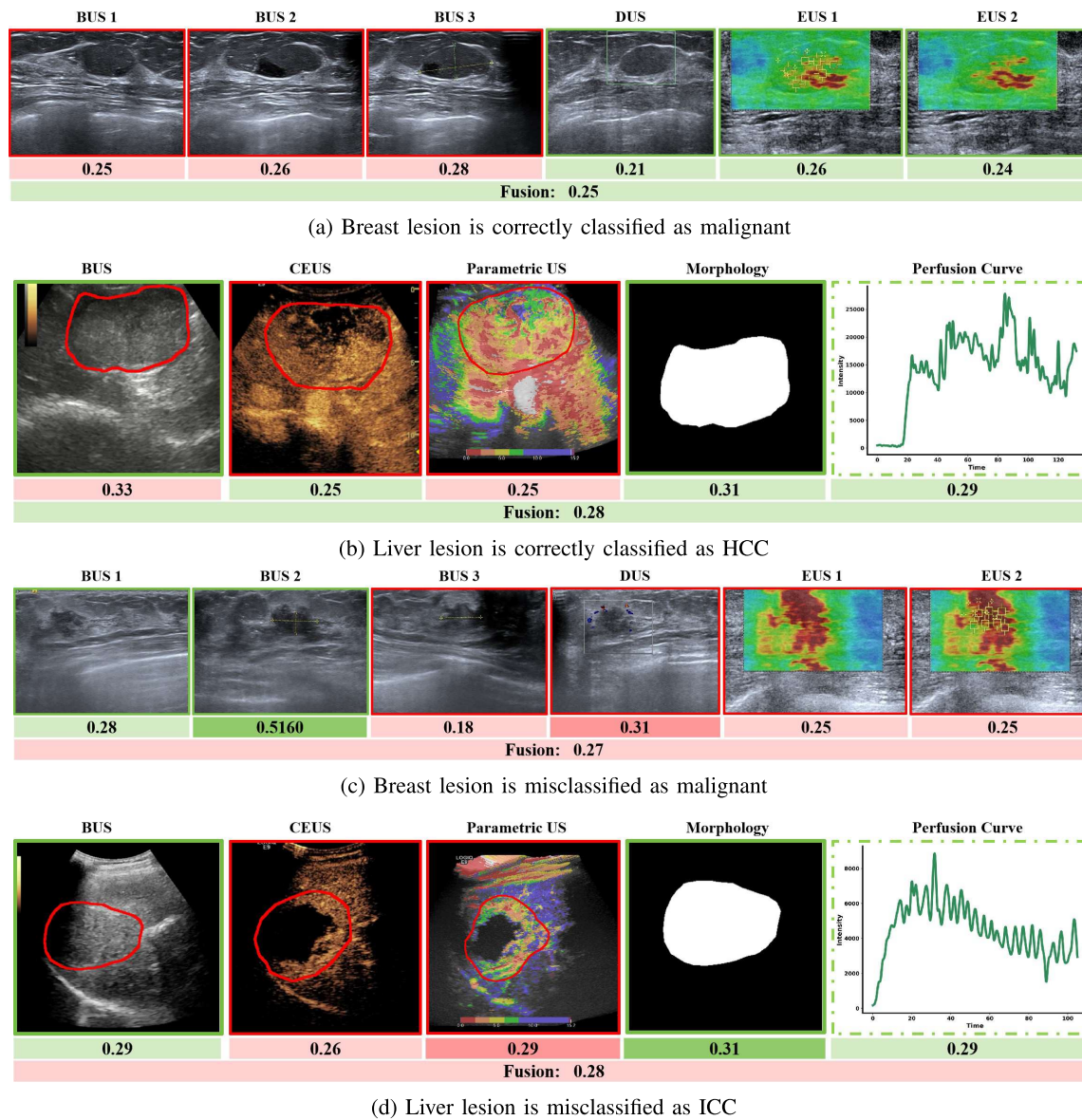


Fig. 7. Qualitative analysis of conflicting modality fusion. (a-b) Successful case analysis; (c-d) Failure case analysis. Red borders indicate incorrect single-modality predictions, green borders indicate correct single-modality predictions, and the numeric values represent the predicted uncertainty scores.

ICC with moderate and high confidence, respectively, while CEUS, Morphology, and Perfusion Curve modalities consistently indicate HCC, driven by characteristic arterial-phase hyperenhancement and washout dynamics. In both cases, the proposed TMUF framework identifies semantic discrepancies across modalities by jointly modeling modality-specific uncertainty and inter-modal disagreement. These examples demonstrate TMUF's robustness in handling cross-modality conflicts and provide interpretable evidence of how modality interactions influence the final decision.

2) Failure Cases Analysis: While the TMUF framework effectively resolves many cross-modality conflicts, it may still struggle when the input data itself contains inherently misleading or cross-category imaging patterns. To further understand such limitations, Fig. 7(c-d) illustrates two representative

failure cases from both breast and liver datasets, where the fusion model failed to produce correct predictions.

In Fig. 7(c), Benign breast lesion is misclassified as malignant. Several BUS, DUS, and EUS views yield high-confidence malignant predictions. This misjudgment is primarily attributed to imaging patterns typically associated with malignancy, most notably the irregular lesion margin in the BUS3 view and the markedly elevated tissue stiffness in the EUS elasticity maps. In Fig. 7(d), HCC is misclassified as ICC. The fusion outcome is dominated by the Parametric US modality, despite correct predictions from BUS, Morphology, and Perfusion Curve. Both CEUS (0.26) and Parametric US (0.29) predict ICC with high confidence. The Parametric US image shows heterogeneous enhancement with irregular peripheral hyperintensity and central hypointensity, a pattern more consistent with ICC in the training distribution.

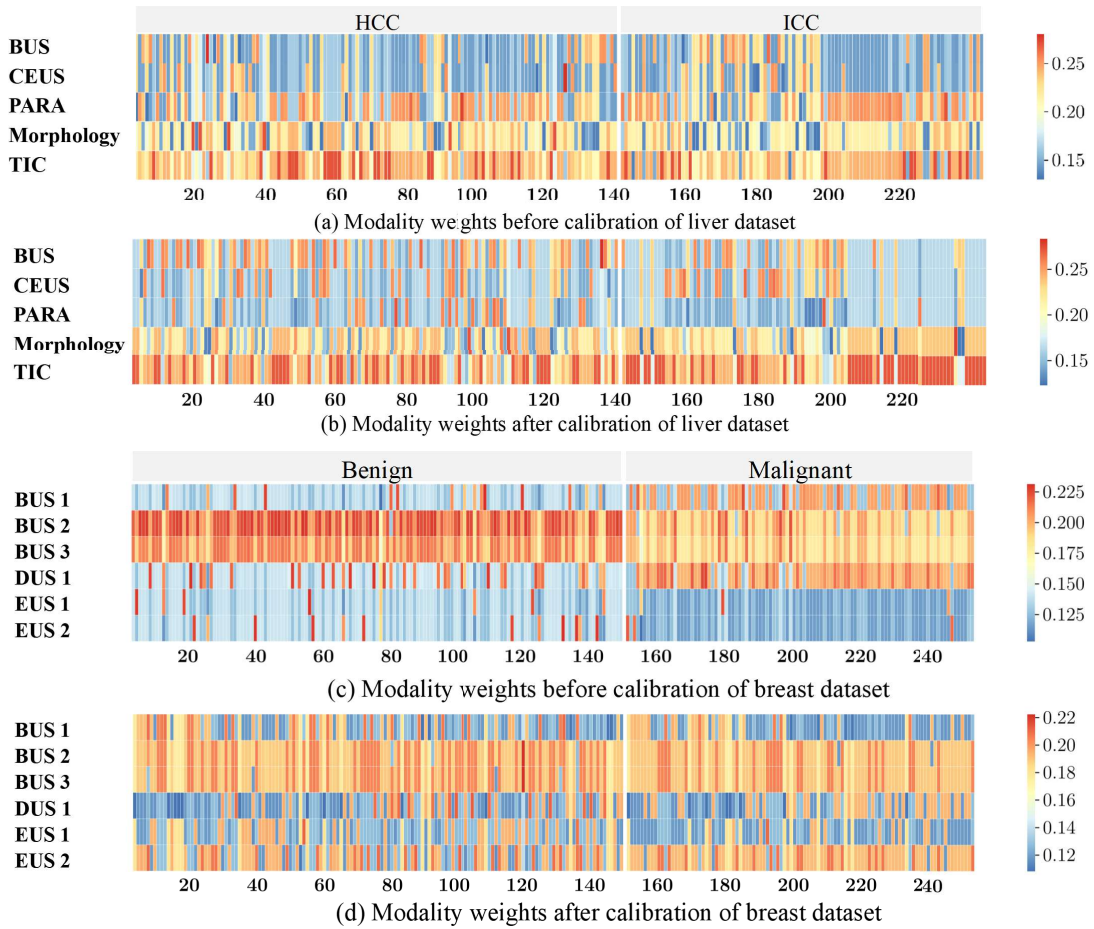


Fig. 8. Modality-wise fusion weights before and after uncertainty calibration for liver and breast ultrasound datasets. After calibration, diagnostically informative modalities such as TIC (liver) and BUS (breast) are assigned higher weights, reflecting improved reliability estimation.

Across these cases, we observe that when the collected dataset inherently contains cross-category imaging patterns—for example, benign lesions exhibiting malignant-like stiffness profiles or ICC lesions displaying HCC-like enhancement curves—the proposed conflict-aware fusion mechanism can still be challenged. These findings suggest that further robustness could be achieved by expanding the dataset scale and diversity to better capture rare or ambiguous feature combinations, thereby enabling the model to learn more discriminative and generalizable representations for challenging cases.

B. Qualitative Analysis of Modality-Wise Fusion Weights

To evaluate the effect of uncertainty calibration, we visualize modality-wise fusion weights before and after calibration for both liver and breast ultrasound datasets, as shown in Fig. 8.

In the liver cancer classification task, the modality weights before calibration (Fig. 8a) are relatively scattered, reflecting inconsistencies in raw fusion signals. After applying our uncertainty calibration strategy (Fig. 8b), the fusion weights shift noticeably toward the TIC and Morphology modalities. Among them, TIC that captures dynamic contrast enhancement curves demonstrates the most consistent and dominant

weighting across both HCC and ICC cases. This highlights the unique diagnostic value of perfusion dynamics in differentiating vascular-rich HCC from fibrotic or necrotic ICC lesions. In contrast, BUS and CEUS, although informative, contribute less consistently due to their higher variability across acquisition planes and imaging conditions, especially in ICC cases where modality discrepancy is more pronounced.

For breast lesion classification, pre-calibration weights (Fig. 8c) are biased toward BUS views, while DUS and EUS show more fluctuation. After calibration (Fig. 8d), fusion weights become more concentrated and reflect modality-specific reliability. Notably, views with unstable performance—such as certain DUS and EUS images—are down-weighted, improving fusion robustness.

These results validate that TMUF effectively leverages uncertainty calibration to emphasize diagnostically informative modalities like TIC, while suppressing noisy or less reliable sources. This adaptive weighting mechanism enhances both the robustness and interpretability of multi-modal ultrasound fusion.

C. Limitation and Future Work

Although TMUF demonstrates consistent diagnostic performance across both public and private datasets, the current

evaluation does not explicitly assess generalization under cross-institutional or cross-device conditions. To address this, future work will explore advanced domain adaptation and generalization strategies, including: 1) adversarial domain adaptation to align feature distributions across domains through discriminative supervision; and 2) causality-inspired learning to capture stable causal factors rather than spurious correlations. In addition, we plan to adopt unseen-center validation protocols (e.g., leave-one-center-out) to provide a more rigorous assessment of generalizability.

Another limitation lies in the lack of explicit interpretability mechanisms. As clinical deployment increasingly demands model transparency, it is a promising direction to integrate explainable AI (XAI) techniques to enhance interpretability of our TMUF. For example, We could explore SHAP (SHapley Additive exPlanations) or Partial Dependency Analysis (PDA) to quantify the contribution of each modality or feature to the final decision [67], [68]. These tools are expected to provide actionable insights into model behavior, thereby improving transparency, usability, and clinician trust.

VI. CONCLUSION

This paper presents a Trustworthy Multi-modal Ultrasound Fusion (TMUF) framework that calibrates modality-wise uncertainty and resolves prediction conflicts through credibility-aware fusion. By aligning uncertainty with predictive correctness and adaptively weighting reliable modalities, TMUF improves diagnostic robustness. Extensive experiments on breast and liver CEUS datasets demonstrate that TMUF outperforms existing fusion methods across all metrics. These results validate the effectiveness of combining uncertainty calibration and dynamic conflict resolution for reliable and interpretable multi-modal diagnosis.

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